A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-51213139})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1-\sigma_x)^2 I' + \mathrm{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 2233 & 54 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,2233 + \mathbf{Z}\,(54 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 12803285 = 0$, of discriminant -51213139. Class group invariants $(75 \quad 5 \quad 5)$, class number 1875; the three stored generator ideals (HNF) are $\begin{pmatrix} 5 & 4 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 843y^3 + (-6s + 3)\, y^2 + 1255594y + (6054s - 3027)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1686*y^8 + 3221837*y^6 - 1656013233*y^4 + 646383262318*y^2 +
469252113896331
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right) s, \qquad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21641357825712 = 2^4 \cdot 3^2 \cdot 29^2 \cdot 178700603$:

$$\begin{pmatrix}
31093904922 & 29383305072 & 24897882398 & 19532123808 & 12355677 & 7402462968 & 1666566853 & 334387905 & 16443131492 & 3241511106 \\
0 & 6 & 5 & 0 & 3 & 3 & 4 & 4 & 5 & 5 \\
0 & 0 & 29 & 0 & 0 & 11 & 20 & 0 & 27 & 5 \\
0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 549 factors with signed exponents of absolute value up to 21397235916377340125450; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 843y^3 + (-6s + 3)\, y^2 + 1255594y + (6054s - 3027)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1686*y^8 + 3221837*y^6 - 1656013233*y^4 + 646383262318*y^2 +
469252113896331
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + (-\frac{3208}{184267035634793})\, s, \qquad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21641357825712 = 2^4 \cdot 3^2 \cdot 29^2 \cdot 178700603$:

$$\begin{pmatrix} 31093904922 & 29383305072 & 24897882398 & 19532123808 & 12355677 & 7402462968 & 1666566853 & 334387905 & 16443131492 & 3241511106 \\ 0 & 6 & 5 & 0 & 3 & 3 & 4 & 4 & 5 & 5 \\ 0 & 0 & 29 & 0 & 0 & 11 & 20 & 0 & 27 & 5 \\ 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 559 factors with signed exponents of absolute value up to 30873431389286011407878; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 843y^3 + (-6s + 3)\, y^2 + 1255594y + (6054s - 3027)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1686*y^8 + 3221837*y^6 - 1656013233*y^4 + 646383262318*y^2 +
469252113896331
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + \left(\frac{50028}{32402005006253125}\right) s, \qquad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $65931043733836 = 2^2 \cdot 29^2 \cdot 19599002299$:

$$\begin{pmatrix}
1136742133342 & 899122946660 & 1136503199418 & 1022028690315 & 333933737358 & 417891789390 & 905292473326 & 569576446612 & 100004718107 & 10943385171 \\
0 & 58 & 52 & 27 & 27 & 6 & 13 & 7 & 37 & 46 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 547 factors with signed exponents of absolute value up to 3187150578186999 6976220; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s - 3108)\, y^3 + (-38s - 70036)\, y^2 + (-427s - 187823)\, y + (-1230s + 968930)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 6201*y^8 - 115242*y^7 + 22650424*y^6 + 1412022498*y^5
+ 35490927781*y^4 + 467314056852*y^3 + 3430948274295*y^2 + 13084669
240100*y + 20307723437500
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right) s, \qquad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2973038062500 = 2^2 \cdot 3^2 \cdot 5^6 \cdot 11^4 \cdot 19^2$:

$$\begin{pmatrix}
344850 & 219450 & 143275 & 292770 & 63785 & 284884 & 289272 & 91371 & 80991 & 59544 \\
0 & 6270 & 3135 & 1430 & 885 & 1 & 2199 & 4294 & 971 & 5079 \\
0 & 0 & 55 & 0 & 50 & 31 & 54 & 52 & 9 & 21 \\
0 & 0 & 0 & 5 & 0 & 1 & 4 & 3 & 1 & 1 \\
0 & 0 & 0 & 0 & 5 & 0 & 3 & 2 & 2 & 3 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 200 factors with signed exponents of absolute value up to 2161798005; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s - 3108)\, y^3 + (-38s - 70036)\, y^2 + (-427s - 187823)\, y + (-1230s + 968930)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 6201*y^8 - 115242*y^7 + 22650424*y^6 + 1412022498*y^5
+ 35490927781*y^4 + 467314056852*y^3 + 3430948274295*y^2 + 13084669
240100*y + 20307723437500
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + (-\frac{3208}{184267035634793})\, s, \qquad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $18403652224655 = 5 \cdot 41 \cdot 2791 \cdot 32165501$:

$$\begin{pmatrix} 18403652224655 & 4325831784230 & 10559134755395 & 4788656683339 & 2023752516102 & 386328830085 & 2895444247320 & 10834469962410 & 12723026731928 & 16815974437789 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 205 factors with signed exponents of absolute value up to 1694470041; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s - 3108)\, y^3 + (-38s - 70036)\, y^2 + (-427s - 187823)\, y + (-1230s + 968930)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 6201*y^8 - 115242*y^7 + 22650424*y^6 + 1412022498*y^5
+ 35490927781*y^4 + 467314056852*y^3 + 3430948274295*y^2 + 13084669
240100*y + 20307723437500
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + (\frac{50028}{32402005006253125})\, s, \qquad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $50090564129296 = 2^4 \cdot 3727 \cdot 14153 \cdot 59351$:

$$\begin{pmatrix} 6261320516162 & 4885507525016 & 335993844314 & 222591634233 & 3384391214741 & 5603543033324 & 1559613696108 & 155628573063 & 4143617540797 & 794667437541 \\ 0 & 2 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 200 factors with signed exponents of absolute value up to 4141560257; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s - 3325)\, y^3 + (16s - 43464)\, y^2 + (-77s + 273390)\, y + (60s + 973900)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 6645*y^8 - 73614*y^7 + 24576112*y^6 + 699498510*y^5 +
1372949137*y^4 - 60244085592*y^3 + 90567820395*y^2 + 414148101700*y
+ 994631470000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad -1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right) s, \qquad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9100531118475 = 3^2 \cdot 5^2 \cdot 19^2 \cdot 2039 \cdot 54949$:

$$\begin{pmatrix} 159658440675 & 43546671824 & 90991179348 & 66415675607 & 114167891859 & 8699385398 & 59416791670 & 71546503190 & 135157323602 & 95609813899 \\ 0 & 19 & 0 & 16 & 3 & 17 & 6 & 11 & 3 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 222 factors with signed exponents of absolute value up to 87432026487069863046; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s - 3325)\, y^3 + (16s - 43464)\, y^2 + (-77s + 273390)\, y + (60s + 973900)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 6645*y^8 - 73614*y^7 + 24576112*y^6 + 699498510*y^5 +
1372949137*y^4 - 60244085592*y^3 + 90567820395*y^2 + 414148101700*y
+ 994631470000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad -1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + \left(-\frac{3208}{184267035634793}\right) s, \qquad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $971142760000 = 2^6 \cdot 5^4 \cdot 7^3 \cdot 70783$:

$$\begin{pmatrix} 8670917500 & 4994192588 & 1482571430 & 1371629620 & 1639388807 & 2152990606 & 6558211230 & 7908670963 & 6997112259 & 1914505825 \\ 0 & 14 & 2 & 12 & 6 & 11 & 8 & 4 & 12 & 6 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 3 & 3 & 2 & 2 & 3 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 222 factors with signed exponents of absolute value up to 22900082111962766400; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 3 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s - 3325)\, y^3 + (16s - 43464)\, y^2 + (-77s + 273390)\, y + (60s + 973900)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 6645*y^8 - 73614*y^7 + 24576112*y^6 + 699498510*y^5 +
1372949137*y^4 - 60244085592*y^3 + 90567820395*y^2 + 414148101700*y
+ 994631470000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad -1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + \left(\frac{50028}{32402005006253125}\right) s, \qquad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $24339292128000 = 2^8 \cdot 3^6 \cdot 5^3 \cdot 1043351$:

$$\begin{pmatrix}
3756063600 & 1306443960 & 2092227624 & 404417550 & 2681935689 & 3266162010 & 1171144293 & 1326581216 & 24929871 & 585551978 \\
0 & 180 & 135 & 153 & 34 & 49 & 109 & 33 & 34 & 24 \\
0 & 0 & 3 & 0 & 1 & 0 & 0 & 1 & 2 & 0 \\
0 & 0 & 0 & 12 & 11 & 11 & 11 & 7 & 7 & 6 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 230 factors with signed exponents of absolute value up to 20500592825989754341 8; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2403y^3 + (-40s + 20)\,y^2 - 3318103y + (7556s - 3778)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 4806*y^8 - 861797*y^6 + 4538452582*y^4 + 3270477952929*y^2 +
730979677478476
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 0 \quad -1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right) s, \qquad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $123319245609877 = 107 \cdot 1493 \cdot 17579 \cdot 43913$:

$$\begin{pmatrix}
123319245609877 & 95352786100400 & 58353283607816 & 51403374606290 & 67807647851331 & 99771014603640 & 110865057524544 & 48005829327421 & 56508900433645 & 22348932069708 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 593 factors with signed exponents of absolute value up to 145379391786; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a+b$, column e_2

Prescribed character vector $(1\quad 1\quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2403y^3 + (-40s + 20)\,y^2 - 3318103y + (7556s - 3778)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 4806*y^8 - 861797*y^6 + 4538452582*y^4 + 3270477952929*y^2 +
730979677478476
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1\quad 0\quad -1\quad -1\quad -1\quad 0\quad 0\quad 0\quad 0\quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + \left(-\frac{3208}{184267035634793}\right) s, \qquad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $152785253526336 = 2^6 \cdot 3^4 \cdot 349 \cdot 84448321$:

$$\begin{pmatrix}
353669568348 & 202585852026 & 297174818703 & 131798455926 & 316398859611 & 248791868016 & 143126282578 & 71985663614 & 132592436906 & 55695810461 \\
0 & 6 & 0 & 0 & 1 & 2 & 0 & 4 & 3 & 5 \\
0 & 0 & 3 & 0 & 2 & 0 & 1 & 1 & 1 & 2 \\
0 & 0 & 0 & 6 & 2 & 2 & 1 & 5 & 2 & 3 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 610 factors with signed exponents of absolute value up to 813188462727; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2\quad 3\quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a+b$, column e_3

Prescribed character vector $(1\quad 1\quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2403y^3 + (-40s + 20)\,y^2 - 3318103y + (7556s - 3778)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 4806*y^8 - 861797*y^6 + 4538452582*y^4 + 3270477952929*y^2 +
730979677478476
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 0 \quad -1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + (\frac{50028}{32402005006253125})\,s, \qquad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7081532492868 = 2^2 \cdot 3^2 \cdot 31^2 \cdot 11743 \cdot 17431$:

$$\begin{pmatrix} 38072755338 & 21737529375 & 17776367461 & 7543254282 & 37315888066 & 15227899647 & 7445854092 & 10117071547 & 2965944340 & 17983903359 \\ 0 & 93 & 44 & 23 & 80 & 59 & 38 & 85 & 40 & 82 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 603 factors with signed exponents of absolute value up to 578035298114; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 - 858y^3 + (18s - 23339)\,y^2 + (18s - 1534266)\,y + (-12420s - 9685584)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 1712*y^8 - 43228*y^7 - 2239030*y^6 + 26787720*y^5 + 7
364105347*y^4 + 96516078642*y^3 - 2914292763072*y^2 + 24014800896696
*y + 2068919484648336
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0\quad 1\quad 1\quad 1\quad 0\quad -1\quad 0\quad 0\quad 0\quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right) s, \qquad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4058422565076 = 2^2 \cdot 3^2 \cdot 11^2 \cdot 931685621$:

$$\begin{pmatrix}
61491250986 & 23087792937 & 23920533138 & 20242310435 & 30102312863 & 33156826612 & 58625072829 & 35413614715 & 25177991972 & 49946227477 \\
0 & 33 & 7 & 6 & 30 & 12 & 12 & 3 & 15 & 17 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 437 factors with signed exponents of absolute value up to 556703612553; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4\quad 1\quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1\quad 0\quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 - 858y^3 + (18s - 23339)\, y^2 + (18s - 1534266)\, y + (-12420s - 9685584)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 1712*y^8 - 43228*y^7 - 2239030*y^6 + 26787720*y^5 + 7
364105347*y^4 + 96516078642*y^3 - 2914292763072*y^2 + 24014800896696
*y + 2068919484648336
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0\quad 1\quad 1\quad 1\quad 0\quad -1\quad 0\quad 0\quad 0\quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + (-\frac{3208}{184267035634793})\, s, \qquad J = \begin{pmatrix} 713 & 705 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2132968950288 = 2^4 \cdot 3^2 \cdot 11 \cdot 25939 \cdot 51913$:

$$\begin{pmatrix}
177747412524 & 136217226853 & 91605633517 & 147112646442 & 156275829024 & 36716626989 & 133151893264 & 168662456848 & 77417134960 & 79758755746 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 1 & 2 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 434 factors with signed exponents of absolute value up to 694224850486; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 4 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 - 858y^3 + (18s - 23339)\, y^2 + (18s - 1534266)\, y + (-12420s - 9685584)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 1712*y^8 - 43228*y^7 - 2239030*y^6 + 26787720*y^5 + 7
364105347*y^4 + 96516078642*y^3 - 2914292763072*y^2 + 24014800896696
*y + 2068919484648336
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + (\frac{50028}{32402005006253125})\, s, \qquad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1309789099076 = 2^2 \cdot 3229 \cdot 101408261$:

$$\begin{pmatrix} 654894549538 & 559463377670 & 34851531669 & 49392909260 & 124295093515 & 479465083738 & 518287054223 & 271375124220 & 563443638587 & 495944555624 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 445 factors with signed exponents of absolute value up to 1006797776379; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2533y^3 + (-12s + 6)\, y^2 + 5184355y + (20100s - 10050)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 5066*y^8 + 16784799*y^6 + 28107615434*y^4 + 20701232202625*y^
2 + 5172655071847500
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 1 \quad -2 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{20608471}{5047204832278763} + \left(\frac{17968}{55519253155066393}\right) s, \qquad J = \begin{pmatrix} 2233 & 54 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2233$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1587485089755 = 3^4 \cdot 5 \cdot 59 \cdot 181 \cdot 367049$:

$$\begin{pmatrix}58795744065 & 2475000921 & 1251773943 & 9878011400 & 53443592271 & 48938037564 & 36762774567 & 58039699757 & 58019424512 & 36227444774\\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 3 & 1 & 2 & 0 & 0 & 1 & 1 & 2\\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 2\\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 1 & 1 & 1\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1\end{pmatrix}$$

The element t is stored in compact form as a product of 243 factors with signed exponents of absolute value up to 402461500; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2533y^3 + (-12s + 6)\, y^2 + 5184355y + (20100s - 10050)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 5066*y^8 + 16784799*y^6 + 28107615434*y^4 + 20701232202625*y^
2 + 5172655071847500
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 1 \quad -2 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7244433}{184267035634793} + \left(-\frac{3208}{184267035634793}\right) s, \qquad J = \begin{pmatrix}713 & 705\\ 0 & 1\end{pmatrix}.$$

$N(J) = 713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $106972481550400 = 2^6 \cdot 5^2 \cdot 171043 \cdot 390883$:

$$\begin{pmatrix}1337156019380 & 608178837060 & 255421563030 & 71537238502 & 1197318463745 & 17724436893 & 5868104429 & 246923783624 & 714713308995 & 1053415292066\\ 0 & 20 & 18 & 13 & 6 & 14 & 10 & 0 & 18 & 5\\ 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1\end{pmatrix}$$

The element t is stored in compact form as a product of 228 factors with signed exponents of absolute value up to 258157777; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2533y^3 + (-12s + 6)\, y^2 + 5184355y + (20100s - 10050)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 5066*y^8 + 16784799*y^6 + 28107615434*y^4 + 20701232202625*y^
2 + 5172655071847500
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 1 \quad -2 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{3788831}{6480401001250625} + \left(\frac{50028}{32402005006253125}\right) s, \qquad J = \begin{pmatrix} 2005 & 589 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2005$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $44221102976000 = 2^{10} \cdot 5^3 \cdot 173 \cdot 1996979$:

$$\begin{pmatrix}
69095473400 & 43020677392 & 57928537602 & 63924889306 & 28361252058 & 60811770029 & 40899964314 & 48446504660 & 13439243281 & 42401016237 \\
0 & 8 & 6 & 4 & 4 & 2 & 4 & 7 & 7 & 3 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 10 & 0 & 4 & 0 & 0 & 0 & 9 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 223 factors with signed exponents of absolute value up to 162971751; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.